ZOO515P – BIOLOGICAL TECHNIQUES

Practical Exam Notes

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MCQs (Multiple Choice Questions)

1. The ratio of Formalin and water is **1:9.**
2. When magnification is increased, the observable area is **INCREASED.**
3. Field of view **DECREASES** when magnification increases.
4. Formula to convert RMP to RCF: **RCF = (1.118 x 10^-5) x RPM^2**
5. Prokaryotes are different from Eukaryotes as the have **SINGLE CELL WITHOUT NUCLEUS AND MEMBRANE ORGANELLES**
6. Most commonly used preservative is **FORMALIN / FORMALDEHYDE**
7. The machine used to separate mixture by using centrifugal force is called **CENTRIFUGE**.
8. Gram staining is a **DIFFERENTIAL** staining.
9. The part that holds slide is called **STAGE CLIPS**.
10. Correct Volumes for Chromatography solution is Butanol : Acetic Acid : Water = **60 : 15 : 25**
11. In gram staining, the secondary dye or counter stain is **SAFRANIN**.
12. Range of P200 micropipette lies between **20-200** microliters.
13. Which of the following does not belongs to dry preservation? **FORMALIN**
14. By using methylene blue, we can see **NUCLEUS** of the cells.
15. There are three main parts of Camera Lucida. **RING ATTACHMENT, PRISM, MIRROR**.
16. The nature of Methylene Blue is **BASIC & POSITIVE CHARGED**.
17. Which of the following is not suitable for intrinsic structure? **DRY MOUNT**
18. In single channel pipette, when working with viscous solutions of solutions prone to bubbles**, REVER TECHNIQUE** is used.
19. Eukaryotic cells range from **10 – 100** micrometer.
20. In physical method of separation, the component to be separated is subjected between two phases. The one is stationary phase and other phase is **MOBILE**.
21. Which is not the characteristic of Eosin stain? **BASIC IN NATURE**
22. For correct calculations of magnification power, the procedure dependent pairing tool are **EYE-OBJECTIVE** lens.

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SHORT QUESTIONS

1. How do you dry preserve dead animals?

Dry Preservation of Dead Animals

Dry preservation is a method used to preserve dead animals without keeping them in liquid preservatives. The animal is first cleaned and internal organs are removed if necessary. The body is then treated with preservatives such as borax, arsenical soap (historically), or other chemicals to prevent decay and insect damage. After treatment, the specimen is dried and mounted in a natural position. Common dry preservation methods include:

1. ***Taxidermy*** – The skin of the animal is removed, treated with preservatives, and mounted on an artificial body form.
2. ***Stuffing and Mounting*** – Small animals and birds may be stuffed with cotton or other materials and displayed.
3. ***Skeleton Preservation*** – Flesh is removed and the cleaned bones are dried and assembled.
4. ***Insect Pinning*** – Insects are killed, dried, and mounted on pins for museum collections.

2. Write role of Methylene Blue in wet mount.

Methylene blue is used as a staining agent in wet mount preparations. It stains cells and microorganisms, making them more visible under the microscope. The dye increases contrast and helps in observing cellular structures such as the nucleus, cytoplasm, and cell boundaries more clearly.

3. Define Rf (Retardation Factor) value.

Rf (Retardation Factor or Retention Factor) is the ratio of the distance traveled by a substance (solute) to the distance traveled by the solvent front during chromatography.

$$Rf = \frac{Distance\ traveled\ by\ the\ solute}{Distance\ traveled\ by\ the\ solvent\ front}$$

4. How to prepare cheek cell slide?

Preparation of Cheek Cell Slide (Steps):

- ✓ Take a clean glass slide.
- ✓ Gently scrape the inner lining of the cheek with a sterile toothpick or cotton swab.
- ✓ Transfer the collected cheek cells onto the slide.
- ✓ Add a drop of methylene blue stain.
- ✓ Place a coverslip carefully over the specimen to avoid air bubbles.
- ✓ Remove excess stain with blotting paper if necessary.
- ✓ Observe the slide under a microscope, starting with low power and then high power.

5. Write functions of pipette.

Functions of Pipette:

- ✓ Measures small volumes of liquids accurately.
- ✓ Transfers liquids from one container to another.
- ✓ Dispenses precise amounts of reagents or samples.
- ✓ Helps in preparing solutions and mixtures.
- ✓ Reduces contamination during laboratory work.

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6. What chemical is used in preservation of dead animals?

Formalin is the most commonly used chemical for preserving dead animals. It is a solution of formaldehyde in water and is usually used at a concentration of 10% formalin. Formalin prevents decomposition by killing bacteria and fungi and preserving body tissues. It helps maintain the shape, structure, and internal organs of specimens for long-term storage in museums, laboratories, and research collections.

7. How micropipette is calibrated?

A micropipette is calibrated by dispensing a known volume of distilled water onto an analytical balance and measuring its weight. Since the density of water is known, the weight is converted into volume and compared with the set volume of the micropipette. If there is a difference, the micropipette is adjusted until the delivered volume matches the desired volume accurately. This process is repeated several times to ensure precision and accuracy.

8. How vertebrates are preserved?

Preservation of Vertebrates:

- ✓ ***Collection of Specimen:*** Collect the vertebrate specimen carefully without damaging its body parts.
- ✓ ***Cleaning:*** Wash the specimen to remove mud, blood, or other contaminants.
- ✓ ***Injection of Formalin:*** Inject 10% formalin into muscles, body cavities, and internal organs to ensure proper preservation.
- ✓ ***Fixation:*** Place the specimen in a jar containing sufficient 10% formalin so that it is completely submerged.
- ✓ ***Preservation Period:*** Keep the specimen in formalin for several days to allow complete fixation of tissues and prevention of decomposition.
- ✓ ***Washing:*** Remove the specimen and wash it thoroughly with water to eliminate excess formalin.
- ✓ ***Transfer to Alcohol:*** Transfer the specimen to 70% ethanol (alcohol) for long-term preservation and storage.
- ✓ ***Labeling and Storage:*** Attach a label containing the specimen name, collection date, and locality, then store it safely in a sealed container for future study and display.

9. What is relationship between field of view and magnification of microscope?

Field of view is the area of the specimen that can be seen through the microscope at one time. Field of view and magnification have an inverse relationship. When the magnification is increased, the field of view becomes smaller because the microscope focuses on a smaller area in greater detail. Conversely, when the magnification is decreased, the field of view becomes larger, allowing a greater area of the specimen to be observed. Therefore, higher magnification provides more detail but a smaller viewing area, while lower magnification provides less detail but a larger viewing area.

10.Write function of glycerin in preservation.

Glycerin is commonly used in the preservation of biological specimens because it acts as a moisturizing and softening agent. It prevents the tissues from drying out and becoming brittle during long-term storage. Glycerin helps maintain the natural shape, size, and flexibility of preserved specimens by retaining moisture within the tissues. It also reduces shrinkage and cracking, thereby improving the appearance and durability of the specimen. For these reasons, glycerin is often added to preservative solutions used in museums, laboratories, and teaching collections.

11.In microscopy, contrast of image is done by which part of compound microscope?

In a compound microscope, the contrast of the image is controlled by the diaphragm (iris diaphragm). The diaphragm regulates the amount of light passing through the specimen, thereby improving image contrast and making details more clearly visible under the microscope.

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12.Write steps for Hematoxylin staining.

Steps for Hematoxylin Staining:

- ✓ Take a clean microscope slide containing the tissue section or prepared smear.
- ✓ Fix the specimen properly (usually with formalin or alcohol) to preserve tissue structure.
- ✓ Rinse the slide gently with water or buffer to remove excess fixative.
- ✓ Immerse the slide in hematoxylin stain for a specific time so nuclei absorb the dye.
- ✓ Wash the slide in running tap water to remove extra stain from the background.
- ✓ Differentiate (if needed) using acid alcohol to remove excess stain and improve clarity of nuclei.
- ✓ Rinse again with water to stop the differentiation process.
- ✓ Perform bluing step by placing the slide in alkaline water or ammonia water, which changes the stain color to blue and enhances nuclear detail.
- ✓ Dehydrate the slide through alcohol grades (if required).
- ✓ Clear the slide using a clearing agent like xylene.
- ✓ Mount the specimen with a mounting medium and cover slip for microscopic examination.

13.Differentiate between fixative and preservative.

Difference between Fixative and Preservative:

Fixative: A fixative is a chemical agent used immediately after collection of a specimen to stop decomposition and maintain the natural structure of tissues as close as possible to the living state. It works by rapidly killing cells, hardening tissues, and preventing enzymatic breakdown and bacterial activity. Fixation is the first step in specimen preparation, commonly using chemicals like 10% formalin.

Preservative: A preservative is a chemical used after fixation for long-term storage of specimens. It prevents decay, microbial growth, and drying of tissues over a longer period. Preservatives help maintain the physical appearance and stability of specimens for teaching, research, and museum display. Common preservatives include 70% alcohol, glycerin, and formalin solutions**.**

14.Why preservation of vertebrates is important?

Importance of Preservation of Vertebrates:

- Helps in the long-term storage of vertebrate specimens by preventing decomposition and maintaining their structural integrity.
- Provides valuable material for teaching and practical learning in biology, zoology, and anatomy laboratories.
- Assists researchers in studying external and internal anatomy in detail without the need for fresh specimens.
- Supports identification, classification, and comparison of different vertebrate species.
- Preserves rare, endangered, or extinct species for scientific record and future reference.
- Plays an important role in museum collections, where specimens are displayed for education and public awareness.
- Contributes to the study of biodiversity and evolutionary relationships among organisms.

15.Why we use formalin and alcohol for preservation of animals?

Formalin and alcohol are commonly used in animal preservation because they effectively prevent decomposition and maintain tissue structure for long periods.

Formalin (10% formaldehyde solution) is used for fixation. It kills microorganisms, stops enzymatic activity, and preserves the natural structure of tissues by hardening them.
Alcohol (usually 70% ethanol) is used for long-term preservation. It helps in removing water from tissues, preventing microbial growth, and reducing decay.

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16.Name the parts of a pipette.

Parts of a Pipette:

- ***Bulb / Rubber filler (pipette bulb):*** Used to suck and release liquid safely.
- ***Suction tube (barrel):*** Main body through which liquid is drawn.
- ***Graduations (scale marks):*** Markings used to measure volume accurately.
- ***Tip (nozzle):*** Lower narrow end through which liquid is dispensed.
- ***Mouthpiece (top opening):*** Upper end where suction is applied in some types of pipettes.

17.Which parts of microscope are used in visualization of Human Hair?

- ***Eyepiece (ocular lens):*** Used to view the magnified image of the hair specimen.
- ***Objective lenses:*** Especially low and high power objectives are used to magnify the hair structure.
- ***Stage:*** Holds the slide containing the human hair specimen.
- ***Stage clips:*** Keep the slide fixed in position during observation.
- ***Light source (illuminator/mirror):*** Provides light to make the hair visible.
- ***Diaphragm:*** Controls the amount of light passing through the specimen to improve clarity and contrast.
- ***Coarse and fine adjustment knobs:*** Used to focus the image of the hair clearly.

18.Why fixatives are used for preservation of animals?

Fixatives are used for preservation of animals because they prevent decomposition and maintain the original structure of tissues as close as possible to the living state. After an animal dies, enzymes and bacteria quickly start breaking down its body. Fixatives such as 10% formalin stop this process by killing microorganisms and inactivating enzymes. They also harden and stabilize proteins in tissues, preventing shrinkage, distortion, and decay. As a result, the specimen retains its shape, cellular structure, and anatomical details, making it suitable for long-term preservation in laboratories, museums, and educational collections.

19.Name the parts of microscope that are used to visualize feathers.

- ***Eyepiece (ocular lens):*** Used to view the magnified image of feather structures.
- ***Objective lenses:*** Provide different magnifications to observe fine details of feather barbules and shafts.
- ***Stage:*** Holds the slide containing the feather specimen.
- ***Stage clips:*** Secure the slide in place during observation.
- ***Light source / mirror:*** Illuminates the feather for clear visibility.
- ***Diaphragm:*** Controls light intensity to improve contrast of feather structures.
- ***Coarse and fine adjustment knobs:*** Used to focus and obtain a sharp image of feather details.

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LONG QUESTIONS

1. Evaluate the killing jar method as a step of preservation of insects.

Killing Jar Method (Evaluation)

The killing jar method is a common first step in insect preservation used to quickly kill collected insects before mounting or storing them. A glass jar is lined with absorbent material soaked in a killing agent (usually ethyl acetate). Insects placed inside are exposed to toxic vapors and die within minutes without much physical damage.

Advantages

- Quickly kills insects without destroying body parts
- Preserves delicate structures like wings, antennae, and legs
- Useful for field collection
- Reduces insect struggle and damage

Disadvantages

- Chemicals can be toxic to humans
- Not suitable for long-term preservation
- May harden or damage specimens if overexposed

Conclusion

The killing jar method is an important initial preservation step, as it helps keep insects in good condition for further procedures like pinning and mounting.

2. Write steps of sample preparation for gram staining.

Steps of Sample Preparation for Gram Staining

Gram staining requires proper preparation of a bacterial smear before staining. The steps are:

1. ***Cleaning the slide:*** A clean glass slide is taken and properly washed to remove grease or dust.
2. ***Preparation of smear:*** A small drop of sterile water (if solid culture) is placed on the slide, then a small amount of bacterial culture is mixed to form a thin smear.
3. ***Spreading the smear:*** The smear is spread evenly to form a thin, uniform layer.
4. ***Air drying:*** The slide is left to air dry completely so that all moisture evaporates.
5. ***Heat fixing:*** The dried smear is gently passed through a flame 2–3 times. This kills bacteria, fixes them to the slide and preserves cell structure.

3. Write steps of gram staining.

Steps of Gram Staining

Gram staining is a differential staining technique used to classify bacteria into Gram-positive and Gram-negative groups.

1. Primary Staining (Crystal Violet)

The prepared smear is flooded with crystal violet stain for about 1 minute, then rinsed with water. All bacterial cells become purple.

2. Mordant Application (Gram's Iodine)

Gram's iodine is added for 1 minute. It forms a crystal violet–iodine (CV-I) complex, fixing the dye inside the cell wall.

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3. Decolorization (Alcohol or Acetone)
Alcohol/acetone is added for a few seconds. ***Gram-positive bacteria*** retain purple color & ***Gram-negative bacteria*** lose color (become colorless)

4. Counter Staining (Safranin)
Safranin is applied for 1–2 minutes. It stains decolorized cells.

- Gram-positive → **Purple**
- Gram-negative → **Pink/Red**

4. Evaluate if killing jar method if insects is ethical or not.

Ethical Evaluation of Killing Jar Method (Insects)
The killing jar method is widely used in entomology to quickly kill insects for scientific study, but its ethical status depends on how it is viewed.

Arguments Supporting Ethical Use
- Insects have a simple nervous system, so they are generally considered to have limited capacity for suffering.
- The method kills insects quickly, reducing prolonged distress.
- It is used for scientific, educational, and conservation purposes (museum collections, taxonomy).
- Helps in studying biodiversity, which can support environmental protection.

Ethical Concerns
- It involves intentional killing of living organisms.
- Large-scale collection may contribute to reduction in insect populations, especially rare species.
- Some people argue that all living organisms deserve protection regardless of size or complexity.
- Risk of over-collection for non-essential purposes.

5. Write method of mount preservation of butterfly.

Mount Preservation Method of Butterfly
Butterflies are preserved mainly by dry mounting (pinning and spreading method) to maintain their shape and wing patterns for study and display.

Steps of Mounting Butterfly
1. **Killing the specimen**
   The butterfly is first killed using a killing jar (usually ethyl acetate vapors).
2. **Softening (if required)**
   If the specimen becomes dry, it is placed in a moist chamber to soften wings and body.
3. **Pinning the butterfly**
   A fine entomological pin is inserted through the thorax (middle body region).
4. **Spreading the wings**
   Wings are carefully opened and arranged on a spreading board so that forewings are placed at right angles and hindwings are properly aligned
5. **Positioning antennae and legs**
   Antennae and legs are adjusted for a natural appearance.
6. **Drying**
   The mounted butterfly is left to dry for several days to 1 week.
7. **Storage**
   After drying, it is stored in an insect box with naphthalene or silica gel for protection from pests and moisture.

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6. How many insect preservation methods are there? Which method will you prefer and why?

Insect Preservation Methods

There are two main categories of insect preservation methods:

1. Dry Preservation Methods

These keep insects in a dry state for long-term storage.

- Pinning (mounting method)
- Spreading (especially butterflies and moths)
- Card mounting
- Taxidermy (in special cases)

2. Wet Preservation Methods

Insects are stored in liquid preservatives.

- Preservation in 70–80% ethanol (alcohol)
- Formalin solution (for soft-bodied insects/larvae)

Preferred Method

The most commonly preferred method is dry preservation (pinning and mounting method).

Reasons for Preference

- Preserves external body structure clearly for identification
- Ideal for museum collections and taxonomy
- Allows long-term storage without decay if protected properly
- Keeps insects in natural shape and position
- Easy to handle and study under microscope

7. Differentiate between RCF (Relative Centrifugal Force) and RPM (Revolutions Per Minute)?

Difference Between RCF and RPM

Feature	RPM (Revolutions Per Minute)	RCF (Relative Centrifugal Force)
Definition	Number of rotations of a centrifuge rotor per minute	Actual force applied on the sample during centrifugation
Unit	rpm	× g (gravity)
Depends on rotor radius?	No	Yes
Accuracy in experiments	Less accurate for comparing machines	More accurate and reliable
Function	Shows speed of spinning	Shows true sedimentation force on particles
Use in labs	Basic setting of centrifuge	Preferred in scientific protocols

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8. What is the use of centrifugation?

Uses of Centrifugation

- ✓ Used for separation of blood components by spinning blood at high speed, where heavier cells (RBCs, WBCs, platelets) settle at the bottom and plasma/serum remains at the top.
- ✓ Helps in urine analysis by separating sediments (cells, crystals, bacteria) from liquid urine for microscopic examination.
- ✓ Used in cell biology and research to isolate different cell organelles such as nucleus, mitochondria, lysosomes, and ribosomes based on density differences.
- ✓ Important in biochemistry laboratories for purification and separation of proteins, enzymes, and nucleic acids (DNA/RNA).
- ✓ Applied in microbiology to concentrate bacteria, yeast, or other microorganisms from liquid cultures for further study.
- ✓ Used in clinical diagnostic laboratories for preparing samples for tests like liver function tests, kidney function tests, and hematology investigations.
- ✓ Helps in separating solid and liquid components in various biological and chemical mixtures quickly and efficiently.

9. By using Picric Acid, Acetic Acid and Formalin, you are supposed to form a solution for the preservation of invertebrates. What will be procedure? Also name the solution.

Name of the Solution

The preservative solution formed by Picric acid, Acetic acid, and Formalin is called Bouin's Solution (Bouin's Fixative).

Procedure for Preparation of Bouin's Solution

- Take a clean glass container (preferably a reagent bottle).
- Measure and mix the following chemicals in correct proportion:
  - Picric acid (saturated aqueous solution)
  - Formalin (37–40% formaldehyde solution)
  - Glacial acetic acid
- A commonly used ratio is:
  - Picric acid solution: 75 mL
  - Formalin: 25 mL
  - Acetic acid: 5 mL
- Add formalin and acetic acid slowly into picric acid solution while stirring gently.
- Mix well until a uniform yellow solution is formed.
- Store the solution in a tightly closed, labeled bottle.

10. Write role of iodine in gram staining.

Iodine acts as a mordant in Gram staining, meaning it helps fix the primary dye (crystal violet) inside the bacterial cell. After crystal violet is applied, iodine is added to form a crystal violet–iodine (CV–I) complex. This complex is larger and more stable than crystal violet alone. Because of its size and insolubility, it becomes trapped more effectively within the thick peptidoglycan layer of Gram-positive bacteria. During the decolorization step (with alcohol or acetone), Gram-positive cells retain this CV–I complex and remain purple, while Gram-negative cells lose the dye because their thinner peptidoglycan layer and outer membrane cannot hold it. In short, iodine's main role is to increase dye retention by forming a stable complex with crystal violet, which is essential for correctly distinguishing Gram-positive and Gram-negative bacteria.

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11.Write applications of Paper Chromatography.

Applications of Paper Chromatography:

Paper chromatography is a simple and widely used analytical technique for separating and identifying components of a mixture. Its main applications include:

1. ***Separation of plant pigments***
   It is used to separate and identify pigments such as chlorophyll, carotenoids, and xanthophylls from plant leaves.
2. ***Analysis of amino acids***
   It helps in separating amino acids in protein hydrolysates, which is useful in biochemical and medical studies.
3. ***Detection of drugs and pharmaceuticals***
   It is used to check the purity of drugs and to identify active pharmaceutical ingredients in formulations.
4. ***Food quality testing***
   It helps detect artificial colors, preservatives, and additives in food products.
5. ***Clinical and medical analysis***
   It is used for identifying substances like sugars, hormones, and metabolic products in body fluids such as urine.
6. ***Forensic investigations***
   It assists in analyzing inks, dyes, and chemical residues in criminal investigations.

12.Write general steps for preservation of invertebrates.

General Steps for Preservation of Invertebrates:

Preservation of invertebrates is important for study, identification, and long-term storage in laboratories and museums. The general steps include:

1. ***Collection of Specimens:*** Invertebrates are collected carefully from their natural habitat using nets, forceps, or traps depending on the organism.

2. ***Killing the Specimen Humanely:*** Specimens are killed using suitable killing agents such as ethyl alcohol, chloroform, or freezing methods to avoid damage.

3. ***Relaxation (if required):*** Some organisms are treated with relaxing agents so that their body parts remain in natural position for proper identification.

4. ***Fixation:*** The specimen is placed in a fixative solution such as formalin (commonly 4–10%) to prevent decay and preserve body structure.

5. ***Washing:*** After fixation, specimens may be washed with water or alcohol to remove excess fixative.

6. ***Storage in Preservative Solution:*** Specimens are stored in jars containing preservatives like 70% ethanol or formalin solution.

7. ***Labeling:*** Each specimen is properly labeled with its name, date of collection, location, and other relevant details.

8. ***Maintenance of Storage Containers:*** Preservative levels are checked regularly and replaced if necessary to ensure long-term preservation.

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13. Write importance of pipettes in laboratory.

Importance of Pipettes in Laboratory:

Pipettes are essential laboratory instruments used for the accurate measurement and transfer of small volumes of liquids. Their importance includes:

1. ***Accurate Measurement of Liquids:*** Pipettes allow precise measurement of specific liquid volumes, which is crucial for experimental accuracy.

2. ***Transfer of Small Volumes:*** They are used to safely transfer small amounts of chemicals, reagents, or biological samples without spillage.

3. ***Improves Experimental Accuracy:*** By ensuring correct volume delivery, pipettes help produce reliable and reproducible results in experiments.

4. ***Essential in Chemical and Biological Experiments:*** Pipettes are widely used in chemistry, microbiology, and medical laboratories for preparing solutions and performing tests.

5. ***Reduces Contamination Risk:*** Using pipettes minimizes direct contact with chemicals, reducing contamination and exposure risks.

6. ***Supports Serial Dilutions and Titrations:*** They are important in procedures that require precise stepwise dilution or measurement of reagents.

14. You've found Dead Sea star shaped echinoderm from the sea shore. Now, you have to preserve it. Can you guide me in steps that how to preserve it? Also name the procedure.

Procedure Name: Wet Preservation (Alcohol/Preservative Method) of Echinoderms

Steps to Preserve a Dead Sea Star (Echinoderm):

1. ***Collection of Specimen:*** Carefully collect the sea star from the shore without breaking its arms or damaging the body surface.

2. ***Cleaning:*** Rinse the specimen gently with clean seawater or fresh water to remove sand, dirt, and debris attached to its body.

3. ***Relaxation:*** If the body is stiff, place it in seawater for a short time so it becomes flexible and retains a natural shape.

4. ***Fixation:*** Immerse the sea star in a fixative solution such as 4–10% formalin for proper preservation of tissues and structure.

5. ***Washing after Fixation:*** After 24–48 hours, wash the specimen with water to remove excess formalin.

6. ***Storage in Preservative Solution:*** Transfer the specimen into a glass jar containing 70% ethanol or diluted formalin for long-term preservation.

7. ***Positioning:*** Arrange the sea star in a natural star-like shape before complete hardening to maintain its anatomical structure.

8. ***Labeling:*** Label the container with details such as name (Sea star), date of collection, and location.

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9. ***Storage Conditions:*** Keep the jar tightly sealed and stored in a cool, dark place to prevent evaporation and degradation.

15.Write the steps to use microscope.

Steps to Use a Microscope:

1. ***Place the Microscope Properly:*** Set the microscope on a stable, flat surface and ensure it is clean and dry.

2. ***Connect the Light Source:*** Switch on the light or adjust the mirror to allow proper illumination.

3. ***Select the Objective Lens:*** Start with the lowest power objective lens (e.g., 4x or 10x) to locate the specimen easily.

4. ***Place the Slide:*** Put the prepared slide on the stage and secure it with stage clips.

5. ***Adjust the Light:*** Regulate the diaphragm and light intensity so the specimen is clearly visible.

6. ***Focus the Image (Coarse Adjustment):*** Use the coarse adjustment knob to bring the specimen into general focus.

7. ***Sharpen the Image (Fine Adjustment):*** Use the fine adjustment knob to get a clear and sharp image.

8. ***Switch to Higher Magnification (if needed):*** Rotate the nosepiece to higher power objectives and refocus using only the fine adjustment knob.

9. ***Observe Carefully:*** Study the specimen systematically without touching the slide or lenses.

10. ***After Use:*** Turn off the light, remove the slide, clean the lenses with lens paper, and cover the microscope properly.

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